

Describe how the big bang theory models the origin and evolution of the universe and analyse the supporting evidence for the theory

Learning intention

- describe how the big bang theory models the origin and evolution of the universe and analyse the supporting evidence for the theory.

Teach and model

- identifying the different technologies used to collect astronomical data and the types of data collected.
- explaining how each different type of evidence, such as cosmic microwave background radiation, red or blue shift of galaxies.
- examining how stars' light spectra and brightness is used to identify compositional elements of stars, their movements and their distances from Earth.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

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Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.